

HEALTHCONNECT EXPERIENCE LAB

Project Title: Healthconnect Clinic Project Summary
Program: AnalystLab Africa (Data Analytics Track)
Week 6

Project Overview

Project: Improving Patient Appointment Attendance and Healthcare Support Using Data and AI

Central Question: How can HealthConnect Clinic use data and AI to reduce missed appointments and improve the patient support experience?

Planned: Integrate the HealthConnect analytical work with the wider project structure; deepen Week 5 findings; validate important KPIs; identify high-impact risk segments; refine dashboard decision support; document limitations; complete a meaningful cross-track integration; and prepare evidence-based coordination for Week 7 testing and refinement.

Completed:

- Completed advanced Week 6 analysis using the established HealthConnect dataset.
- Validated the 48.5% overall no-show rate and the core Week 5 KPI baseline.
- Created and reviewed Page 4 - Advanced No-Show Analysis.
- Created and reviewed Page 5 - Reminder & Risk Segmentation.
- Derived practical risk segmentation using previous no-show history.
- Examined interactions between previous no-show history, distance, reminder status, appointment type and age/time segments.
- Completed a documented Data Analytics <> Project Management integration.

What Improved from Week 5: Week 5 established baseline performance and single-factor patterns. Week 6 moved beyond baseline EDA into cross-segment analysis, risk-focused interpretation, KPI validation, evidence-based recommendations and project-level integration.

What Was Integrated: The Data Analytics Week 5 output was integrated into Project Management coordination. The output was reviewed and used as an input to the PM integration/dependency view and Week 7 testing preparation.

Collaborated Track: Project Management. The completed evidence-based integration pathway is Data Analytics <> Project Management.

Information Exchanged:

Direction	Information exchanged
Data Analytics → PM	Week 5 appointment attendance/no-show analysis, KPIs, findings, relevant factors, limitations and remaining analytical work.
PM → Data Analytics	Integration context, dependencies, risks/priorities and Week 7 testing/refinement requirements.

What Changed: The analytics findings were converted from a track-level output into a documented PM integration input. The findings now inform dependency tracking and Week 7 testing preparation. Remaining analytical work and data limitations are also carried forward rather than being treated as complete.

Findings / Outcomes

- Overall no-show rate remains 48.5%.
- Previous no-show history showed an increasing observed no-show pattern: 43.5% for 0, 54.8% for 1–2 and 68.8% for 3+ previous no-shows.
- The 21+ km group recorded a 57.8% observed no-show rate.
- No-reminder appointments recorded 51.4% no-show compared with 47.4% among reminder appointments, an observed 4.0 percentage-point gap.
- Follow-up appointments recorded the highest appointment-type no-show rate at 51.2%.
- Booking lead-time data remains an analytical limitation and was not treated as a validated Week 6 finding.

Challenges

- Some variables required careful treatment because of data-quality concerns.
- Not all technical workstream outputs were available for independent verification.
- The project requires a clear distinction between planned integration pathways and completed evidence-based integration.
- Some high-risk segments may have smaller sample sizes and therefore require cautious interpretation.

Decisions

- Retain the Week 5 dashboard pages as the baseline and use additional pages for Week 6 advanced analysis.
- Prioritize previous no-show history, distance, reminder status and appointment type for deeper testing.
- Do not make causal claims from observational patterns.
- Do not treat booking lead time as a validated finding until its data-quality issues are resolved.
- Treat Data Analytics → PM as the completed Week 6 integration pathway and keep other technical pathways pending evidence.

Remaining Issues

- Validate remaining technical workstream outputs before end-to-end testing.
- Resolve or formally track outstanding integration dependencies and issues.
- Continue analytical validation during Week 7 testing.
- Review booking lead-time data quality before downstream predictive use.
- Document whether targeted interventions improve attendance.

Data Analytics Contribution: The Data Analytics contribution was the preparation of the HealthConnect attendance/no-show analysis, advanced segment-level investigation, KPI validation, identification of priority risk areas, evidence-based recommendations, documentation of limitations, dashboard refinement, and provision of analytical evidence to Project Management for Week 7 coordination.

Week 7 Focus

1. Test enhanced reminder strategies for patients with previous no-show history.

2. Investigate targeted support for patients living 21+ km from the clinic.
3. Evaluate reminder coverage and channel performance across relevant appointment segments.
4. Monitor follow-up appointments as a higher-risk segment.
5. Validate whether identified risk patterns remain consistent during testing.
6. Verify remaining technical workstream outputs and dependencies before broader end-to-end testing.

Cross-Track Integration Evidence

What PM gave the Data Analytics track: Project Management provided the wider integration context, dependency/risk considerations, project priorities and the Week 7 testing/refinement focus.

What was actually exchanged: Data Analytics supplied its Week 5 HealthConnect appointment attendance/no-show analysis and its analytical limitations and remaining work. PM supplied integration context and testing coordination requirements.

What changed as a result: The analytics output was incorporated into the PM integration/dependency view and used to support Week 7 preparation.

How the output was used: The findings were used to identify and prioritize analytical considerations for testing and to strengthen evidence-based project coordination. No downstream technical use by Data Science, ML Engineering or Generative AI is claimed without supporting evidence.

Week 6 strengthened the HealthConnect Data Analytics contribution by moving from baseline reporting to advanced risk segmentation, KPI validation and project-level integration. The completed Data Analytics $\diamond$ PM exchange provides a documented integration point and supports a focused Week 7 testing and refinement stage.